

From Length–Dependent to Mass–Dependent Stiffness in the MMU:

$k_i(a) \rightarrow k_i(m)$ and Leptonic Balmer Predictions

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Abstract

In the Methane–Metric Universe (MMU) model, the internal dynamics of a tetrahedral elastic cell are described by stiffnesses $k_2(a)$, $k_3(a)$ and $k_4(a)$ that scale with the internal edge length a . In this paper we close the system by expressing all stiffnesses as explicit functions of mass, using the torsional energy closure $a(m) = \hbar/(mc)$ established in earlier MMU work. This turns the MMU cell into a purely mass–driven object and yields the key scalings $k_2(m) \propto m^3$, $k_3(m) \propto m^3$ and $k_4(m) \propto m^5$. We show how these relations generate a leptonic Balmer scaling for hydrogenic atoms (electron, muon, tau) and provide explicit predictions for the Balmer α line energies and wavelengths. The resulting spectrum reproduces the known electronic and muonic values and extends the MMU framework to hypothetical tauonic atoms. We also highlight how the mass–dependent stiffnesses relate to the recently identified electro–gravity fixed point [11], complementing the spin–gravity (Planck) scale [10].

1 Introduction

The MMU framework models space and matter as a repeating tetrahedral elastic cell with internal modes (w_1, w_2, w_3, w_4) and associated stiffnesses k_2, k_3, k_4 [1, 2, 8]. In earlier work, these stiffnesses were expressed as functions of the internal edge length a ,

$$k_2(a) = \frac{2\alpha\hbar c}{a^3}, \quad k_3(a) = \frac{\hbar c}{a^3}, \quad k_4(a) = \frac{2Gm^2}{a^3}, \quad (1)$$

where α is the fine–structure constant, G the gravitational constant, $\hbar$ the reduced Planck constant and c the speed of light [5, 10]. The conversion of these relations into *mass–only* functions $k_i(m)$ is a crucial step because it turns the MMU cell into a fully mass–driven object.

The torsional energy closure $a(m) = \hbar/(mc)$, originally derived in [8, 9], provides this link between geometry and mass. Once $k_i(a)$ are rewritten as $k_i(m)$, the MMU stiffness hierarchy becomes explicitly mass–ordered, revealing the structure of the electro–gravity and spin–gravity fixed points [10, 11]. This mass dependence naturally feeds into the MMU derivation of hydrogenic spectra across the lepton families.

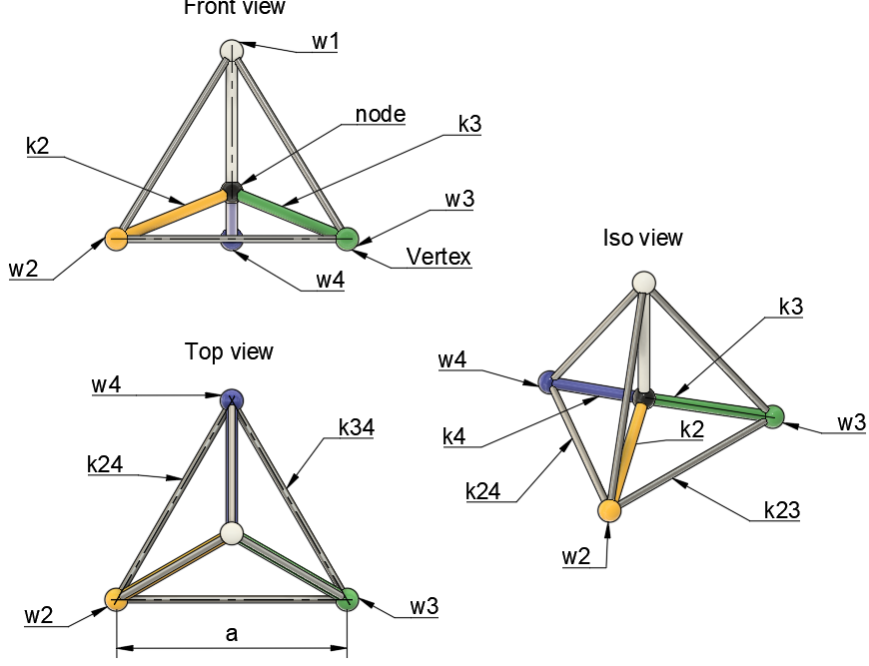


Figure 1: MMU tetrahedral cell with diagonal stiffnesses k_2 , k_3 , k_4 and cross-couplings k_{23} , k_{24} , k_{34} . Front, top and isometric views illustrate how the electric (w_2), torsional (w_3) and volumetric (w_4) directions are connected through the elastic edges of the cell.

2 From $k_i(a)$ to $k_i(m)$

2.1 Energy closure and internal edge length

In the MMU, the minimal torsional excitation of a cell of size a is taken to scale as

$$E_{\text{tors}}(a) \sim \frac{\hbar c}{a}, \quad (2)$$

a relation that underlies the MMU Compton-like quantisation of internal cell size [8]. Equating this to the rest energy of a fermionic excitation of mass m ,

$$E_{\text{tors}}(a) = mc^2, \quad (3)$$

gives the mass-dependent internal edge length

$$a(m) = \frac{\hbar}{mc}. \quad (4)$$

This identifies each MMU cell with the Compton wavelength of its fermionic excitation.

2.2 Mass-dependent stiffnesses

Inserting $a(m) = \hbar/(mc)$ into the length-dependent stiffnesses yields

$$k_2(m) = \frac{2\alpha\hbar c}{a(m)^3} = 2\alpha \frac{m^3 c^4}{\hbar^2}, \quad (5)$$

$$k_3(m) = \frac{\hbar c}{a(m)^3} = \frac{m^3 c^4}{\hbar^2}, \quad (6)$$

$$k_4(m) = \frac{2Gm^2}{a(m)^3} = 2G \frac{m^5 c^3}{\hbar^3}. \quad (7)$$

Thus electric and torsional stiffness grow as m^3 , while gravitational stiffness grows as m^5 . This mass hierarchy explains why the volumetric gravitational mode dominates at large mass and why the equalities $k_3 = k_4$ and $k_2 = k_4$ define fundamental fixed points of the MMU cell. The former gives the Planck mass [10]; the latter defines the newly identified electro-gravity mass m_{24} [11].

3 Leptonic Balmer Scaling in the MMU

3.1 Reduced mass and Rydberg scaling

For a hydrogenic atom with nuclear mass m_p and lepton mass m_ℓ the reduced mass is

$$\mu_\ell = \frac{m_\ell m_p}{m_\ell + m_p}. \quad (8)$$

In the Bohr/Rydberg picture the level energies scale as $E_n \propto -\mu_\ell/n^2$. In the MMU this scaling arises from the combination of $k_2(m_\ell) \propto m_\ell^3$ and $a(m_\ell) \propto 1/m_\ell$, yielding an effective binding energy $E \sim k_2 a^2 \propto m_\ell$. This directly reproduces the reduced-mass scaling.

3.2 Numerical predictions for electron, muon and tau

Using $E_{\alpha,e} = 1.889$ eV for the hydrogen Balmer α transition and standard physical lepton and proton masses, the MMU scaling gives

$$E_{\alpha,\mu} \approx 351 \text{ eV}, \quad (9)$$

$$E_{\alpha,\tau} \approx 2.27 \text{ keV}. \quad (10)$$

With $hc = 1239.84$ eV nm the corresponding wavelengths are

$$\lambda_{\alpha,e} \approx 656.3 \text{ nm}, \quad (11)$$

$$\lambda_{\alpha,\mu} \approx 3.53 \text{ nm}, \quad (12)$$

$$\lambda_{\alpha,\tau} \approx 0.546 \text{ nm}. \quad (13)$$

Table 1 summarises these results.

Table 1: MMU predictions for the Balmer α transition ($3 \rightarrow 2$) in hydrogenic atoms with electron, muon and tau leptons. The electronic value is taken from experiment; the muonic and tauonic values follow from the MMU mass scaling.

Lepton	Reduced mass ratio μ_ℓ/μ_e	$E_{\alpha,\ell}$ [eV]	$\lambda_{\alpha,\ell}$ [nm]
Electron e	1	1.889	656.3
Muon μ	185.9	3.51×10^2	3.53
Tau τ	1202	2.27×10^3	0.546

4 Discussion

By converting the MMU stiffnesses from length-dependent to mass-dependent functions we obtain a compact description of how electric, torsional and gravitational responses scale across the lepton families. The relations $k_2(m) \sim m^3$ and $a(m) \sim 1/m$ directly reproduce the Rydberg reduced-mass law and provide a geometric, mechanical interpretation of the leptonic Balmer series. The mass scaling also underlies the two fundamental MMU fixed points: the spin-gravity (Planck) limit [10] and the newly discovered electro-gravity fixed point [11]. Together they bracket the full range of matter behaviour from atomic spectroscopy to gravitational collapse.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantization. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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